Activity 2: Upgrade the Computer Lab or Keep Old PCs?

Annual Worth (AW) and Rate of Return (IRR) Analysis

Objective.
The Computer Engineering lab is considering whether to keep the existing PCs or replace them with new, energy-efficient machines. You will perform an incremental analysis to evaluate the upgrade.

Assume: All cash flows are in $ per year. Study period is 5 years (the life of the new PCs).

Option 1: Keep Old PCs

- No initial investment (the old PCs are already owned).
- Annual maintenance cost: $4,000 per year.
- Old PCs are less efficient and cause extra electricity cost, but no additional salvage value is considered.

Option 2: Buy New PCs

- Initial cost of new PCs: $30,000 at year 0.
- Annual maintenance cost: $1,000 per year.
- Annual energy savings compared to old PCs: $1,000 per year.
- Salvage value at the end of year 5: $5,000.
- Useful life: 5 years.

Interpretation of savings:
Each year, the new PCs save money compared to the old PCs due to:

- Lower maintenance: $4,000 (old) vs. $1,000 (new) $\Rightarrow$ $3,000 savings
- Lower energy cost: additional $1,000 savings per year

So the total annual savings if we upgrade is: **$4,000 per year** compared to keeping the old PCs.

Tasks

(a) Construct the incremental cash flow (Upgrade − Keep Old).
Fill in the incremental cash flow table (from the point of view of "Upgrade" vs. "Keep Old"):

Year	Incremental Cash Flow (Upgrade − Keep Old)
0	_____
1	_____
2	_____
3	_____
4	_____
5	_____

(b) Estimate the IRR of the upgrade.
Is the internal rate of return (IRR) positive or negative? Based on the sign and magnitude of IRR, would you recommend upgrading?

$$\text{IRR} \approx _____$$

Decision based on IRR: _____

(c) Use Annual Worth (AW) to check your decision.
Assume MARR is 10%. Compute the AW of the incremental cash flow at 10%:

$$AW_{\Delta} = _____$$

Decision based on AW: _____

Answer Key

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General Factor Formulas

$$(A/P, i, n) = \frac{i(1+i)^n}{(1+i)^n - 1}$$

$$(A/F, i, n) = \frac{i}{(1+i)^n - 1}$$

$$(P/F, i, n) = \frac{1}{(1+i)^n}$$

For $i = 10\%$ and $n = 5$:

$$(A/P, 10\%, 5) = 0.2638, \qquad (P/F, 10\%, 5) = \frac{1}{1.1^5} = 0.6209$$

(a) Incremental Cash Flow (Upgrade − Keep Old)

Year	Incremental CF
0	−30,000
1	+4,000
2	+4,000
3	+4,000
4	+4,000
5	+4,000 + 5,000 = 9,000

(b) IRR of the Upgrade

IRR is the value of r that satisfies:

$$-30{,}000 + \sum_{t=1}^{4} \frac{4{,}000}{(1+r)^t} + \frac{9{,}000}{(1+r)^5} = 0$$

Trial: - At $r = 0\%$: PW $= +-30{,}000 + 4{,}000(4) + 9{,}000 = +-30{,}000 + 25{,}000 = -5{,}000$ - At negative rates, PW increases.

Computed using solver:

$$\text{IRR} \approx -5.1\%$$

Decision: IRR is negative $\Rightarrow$ upgrade is not attractive.

(c) AW of the Incremental Cash Flow at 10%

Step 1. Compute PW of incremental cash flow

$$PW_\Delta = -30{,}000 + 4{,}000\left(\frac{1}{1.1} + \frac{1}{1.1^2} + \frac{1}{1.1^3} + \frac{1}{1.1^4}\right) + 9{,}000(P/F, 10\%, 5)$$

Compute each term:

$$\frac{1}{1.1} = 0.9091, \quad \frac{1}{1.1^2} = 0.8264, \quad \frac{1}{1.1^3} = 0.7513, \quad \frac{1}{1.1^4} = 0.6830$$

Sum:

$$0.9091 + 0.8264 + 0.7513 + 0.6830 = 3.1698$$

Therefore:

$$4{,}000(3.1698) = 12{,}679.2$$

Salvage term:

$$9{,}000(P/F, 10\%, 5) = 9{,}000(0.6209) = 5{,}588.1$$

Full PW:

$$PW_\Delta = -30{,}000 + 12{,}679.2 + 5{,}588.1$$
$$PW_\Delta = -11{,}732.7$$

Step 2. Convert PW to AW

$$AW_\Delta = PW_\Delta(A/P, 10\%, 5)$$
$$AW_\Delta = -11{,}732.7(0.2638)$$
$$AW_\Delta \approx -3{,}095$$

Decision: Since $AW_\Delta < 0$, the upgrade is not justified. **Keeping the old PCs is preferred.**